

Practice problems #9

1. Suppose the eigenvalues of a 3×3 matrix A are 3, $4/5$, and $3/5$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 1 \\ -5 \end{bmatrix}$, and $\begin{bmatrix} -2 \\ -5 \\ 3 \end{bmatrix}$. Let $x_0 = \begin{bmatrix} -3 \\ -3 \\ 7 \end{bmatrix}$. Find the solution of the equation $x_{k+1} = Ax_k$ for the specified x_0 , and describe what happens as $k \rightarrow \infty$.
2. Let A be a 2×2 matrix with eigenvalues 3 and $1/3$ and corresponding eigenvectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Consider the difference equation $x_{k+1} = Ax_k$ with $x_0 = \begin{bmatrix} 9 \\ 1 \end{bmatrix}$.
 - (a) Compute x_1 *without* finding the matrix A itself.
 - (b) Find a formula for x_k .
3. Suppose the female dogs give birth each year to an average of 1.6 female puppies. Each year, 30% of the puppies survive to become dogs and 80% of the dogs survive. Consider a difference equation $x_{k+1} = Ax_k, k \geq 0$, where $x_k = (P_k, D_k)$, the numbers of female puppies and female dogs in year k .
 - (a) Find the matrix A .
 - (b) Show that population is growing.
 - (c) Fill in the blanks and explain why:
Eventually, both populations will grow at _____% per year. Eventually, there will be _____female puppies for every _____female dogs.

Hints or Answers on page 2.

#1: $x_k = 3^k(-1, 0, 3) + (3/5)^k(-1, -5/2, 3/2) + (4/5)^k(-1, -1/2, 5/2)$.

As $k \rightarrow \infty$, $x_k \rightarrow (-\infty, 0, \infty)$.

#2(a): $x_1 = \begin{bmatrix} 49/3 \\ 41/3 \end{bmatrix}$

#2(b): $x_k = 5(3^k)\mathbf{v}_1 - 4((1/3)^k)\mathbf{v}_2$ for $k \geq 0$.

#3(a): $A = \begin{bmatrix} 0 & 1.6 \\ 0.3 & 0.8 \end{bmatrix}$

#3(b): What is the largest eigenvalue?

#3(c): Ans: 20, 4, 3. Explain why.